File Server, Access and Recovery

Windows Server 2025 / Active Directory / SMB / Group Policy / Backup and Recovery

Personal lab project

Domain: lab.local

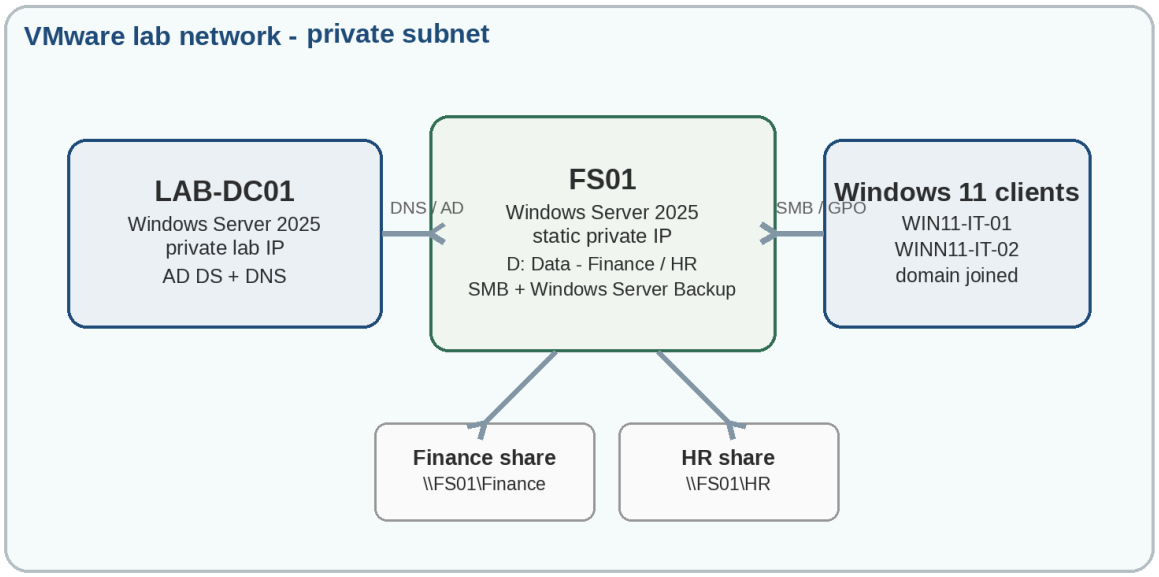
October 2026

This report documents the completed P2 lab as six support-style tickets. The screenshots are from the working lab and are used as evidence for the decisions, configuration and tests described in each section.

Project overview

This project extends my lab.local Active Directory lab with a dedicated member file server, FS01. I worked through six support-style cases: Finance and HR shares, role-based access, automatic drive mapping, an employee transfer, file recovery, and an access-denied case after a group change. Each change was tested from a Windows client as the affected user. I kept the mistakes and failures that mattered because they show how I checked the problem instead of only showing the finished configuration.

Environment: DC01 provides AD DS and DNS. FS01 runs Windows Server 2025 with a static private IP and stores file data on a separate D: data disk. WIN11-IT-01 and WINN11-IT-02 were used for user-side testing. The screenshots are from the lab sessions; the architecture diagram is a simplified illustration.



Lab layout used for the project.

System	Role	Address / purpose
LAB-DC01	Domain controller	Private lab IP - AD DS and DNS
FS01	Member file server	Static private IP - D: data disk, Finance and HR shares
WIN11-IT-01 / WINN11-IT-02	Domain clients	Used to test user access and mapped drives

Access model

Access follows an AGDLP-style structure: users go into Global groups that describe their role, those Global groups are nested into Domain Local groups that describe the resource permission, and only the Domain Local groups are placed on the FS01 folder ACLs. In simple terms: user -> role group -> permission group -> folder. That means a joiner, mover or leaver can usually be handled by changing group membership instead of editing the folder permissions for one person.

User / role group	Resource group	NTFS	SMB share
GG_Finance_Users	DL_FS01_Finance_Modify	Modify	Change
GG_Finance_Reviewers	DL_FS01_Finance_Read	Read and execute	Read
GG_HR_Users	DL_FS01_HR_Modify	Modify	Change

Administrators and SYSTEM retain full control on the server. The effective user access is the combination of SMB share permissions and NTFS permissions; the more restrictive result applies.

Ticket #2001 - Finance share and role-based access

Finance needs modify access, the reviewer needs read-only access, and non-Finance users must be refused.

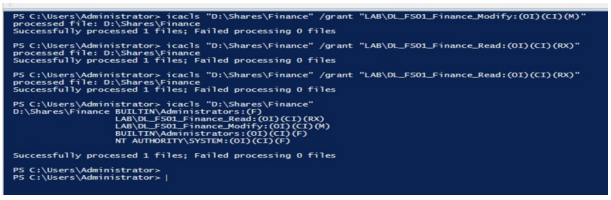
Request: Create a Finance share where finance.test01 can create, edit and delete files, finance.reviewer can read but cannot change files, and users outside Finance cannot open the folder. No user account should be placed directly on the folder ACL.

What I checked: I checked the Finance global groups, their nested domain-local resource groups, the NTFS ACL on D:\Shares\Finance, and the SMB share permissions for \\FS01\Finance.

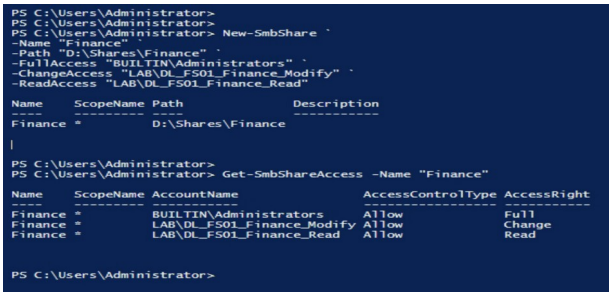
What I did: I granted DL_FS01_Finance_Modify NTFS Modify and SMB Change, and DL_FS01_Finance_Read NTFS Read/Execute and SMB Read. finance.test01 was placed in GG_Finance_Users and finance.reviewer in GG_Finance_Reviewers. Administrators and SYSTEM kept full control.

How I verified it: finance.test01 created and deleted a test file. finance.reviewer could open the document, but a save/replace attempt was refused. hr.test01 and noaccess.test01 both received Access is denied when they tried the Finance UNC path.

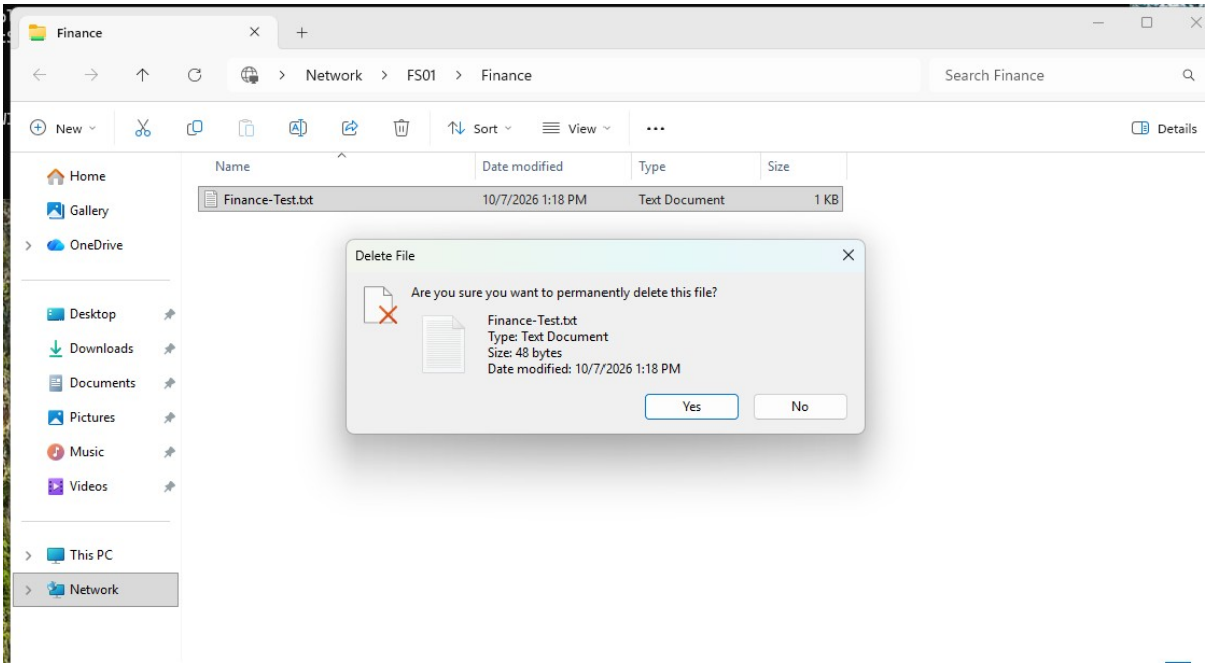
Issue encountered: My first client test used \\FS01\\Finance with an extra backslash between the server and share names. DNS resolution and TCP 445 were working, so the problem was the path, not SMB connectivity or permissions. After correcting it to \\FS01\Finance, the access tests behaved as expected.



Finance NTFS ACL: resource groups receive Modify and Read/Execute.

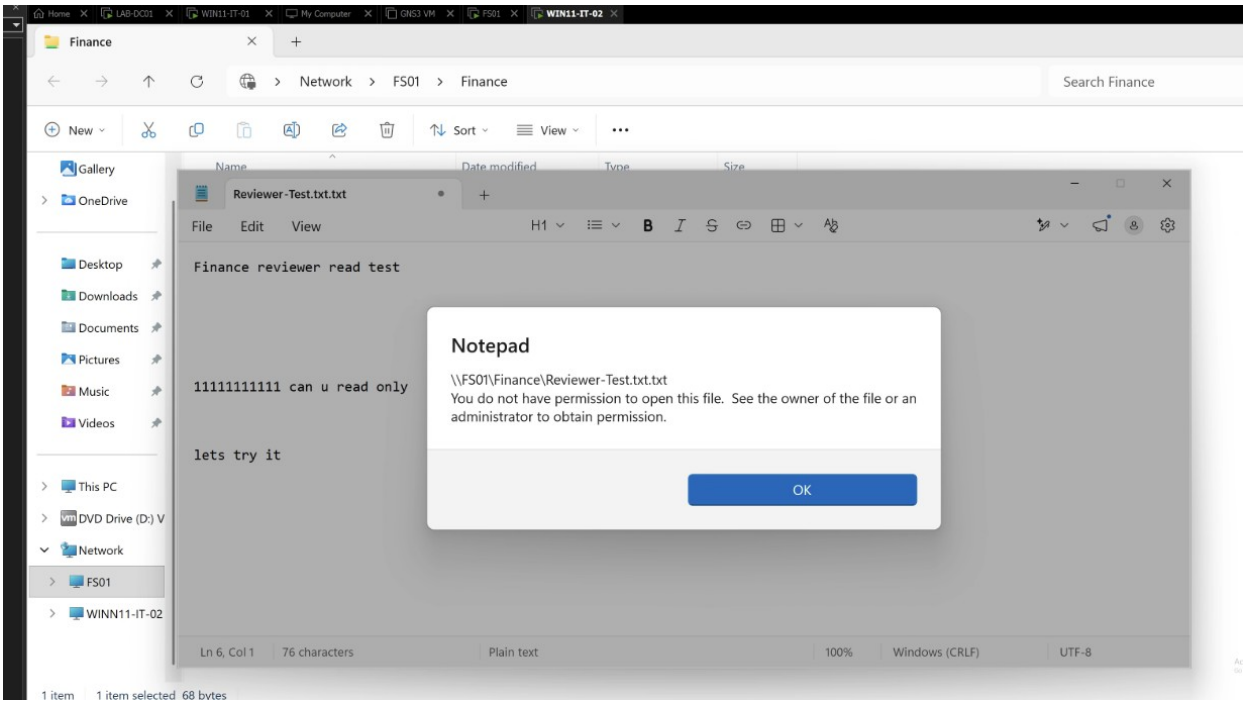


Finance SMB share: Change for the modify group and Read for the reviewer group.

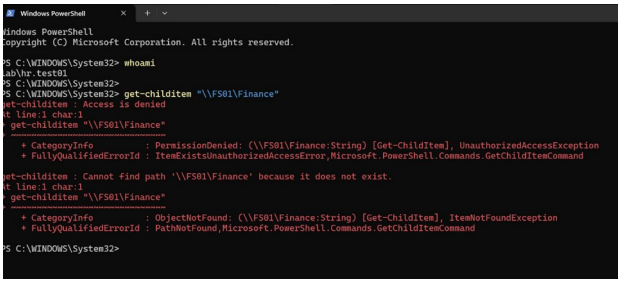


finance.test01 was able to create a test file and reach the delete operation on the Finance share.

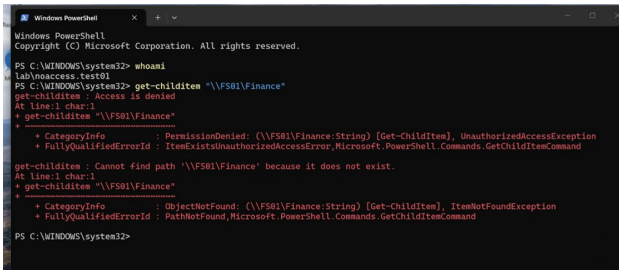
Ticket #2001 - access tests



The Finance reviewer could read the document, but the attempted save/replace was refused.



hr.test01 was refused at \\FS01\Finance.



noaccess.test01 was also refused at \\FS01\Finance.

PowerShell sometimes printed a second "cannot find path" message after Access is denied. The first PermissionDenied / UnauthorizedAccessException is the important result: the share exists, but the user is not authorized to enumerate it.

Ticket #2002 - HR share

HR must have its own share and Finance must not be able to use the HR network path.

Request: Create an HR folder that only HR can change. Finance users must be refused, including when they type the full UNC path directly.

What I checked: I checked the HR global and domain-local groups, the NTFS ACL on D:\Shares\HR and the SMB permissions for \\FS01\HR.

What I did: I created GG_HR_Users and DL_FS01_HR_Modify, added hr.test01 to the global group and nested that group into the domain-local resource group. The HR resource group received NTFS Modify and SMB Change.

How I verified it: hr.test01 created and deleted HR-Test.txt. finance.test01 then typed the direct UNC path \\FS01\HR and received Access is denied. This showed that knowing the full network path did not bypass the HR permissions.

Issue encountered: The first direct-path test had a typo in the server name and returned Cannot find path. I corrected the path and reran the test. Cannot find path proves only that the path is wrong; Access is denied is the permission result I needed.

```
PS C:\Users\Administrator> New-ADGroup GG_HR_Users -GroupScope Global -GroupCategory Security -Path "OU=PortfolioLab,DC=lab,DC=local"
New-ADGroup DL_FS01_HR_Modify -GroupScope DomainLocal -GroupCategory Security -Path "OU=PortfolioLab,DC=lab,DC=local"
Add-ADGroupMember GG_HR_Users -Members hr.test01
Add-ADGroupMember DL_FS01_HR_Modify -Members GG_HR_Users

PS C:\Users\Administrator> "GG_HR_Users","DL_FS01_HR_Modify" | ForEach-Object { "--- $_"; Get-ADGroupMember $_ | Select -Expand SamAccountN
--- GG_HR_Users
hr.test01
--- DL_FS01_HR_Modify
GG_HR_Users

PS C:\Users\Administrator>
```

HR group structure: hr.test01 -> GG_HR_Users -> DL_FS01_HR_Modify.

```
D:\Shares\HR BUILTIN\Administrators:(F)
             LAB\DL_FS01_HR_Modify:(OI)(CI)(M)
             BUILTIN\Administrators:(OI)(CI)(F)
             NT AUTHORITY\SYSTEM:(OI)(CI)(F)

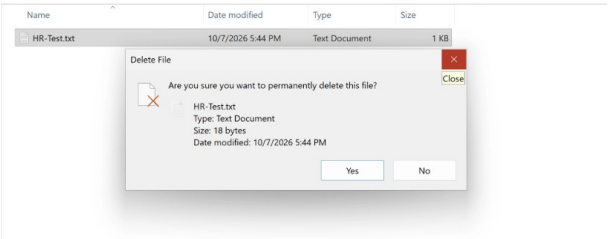
Successfully processed 1 files; Failed processing 0 files

AccessControlType : Allow
AccessRight       : Full
AccountName       : BUILTIN\Administrators
Name              : HR
ScopeName         : *
PSComputerName    :

AccessControlType : Allow
AccessRight       : Change
AccountName       : LAB\DL_FS01_HR_Modify
Name              : HR
ScopeName         : *
PSComputerName    :


PS C:\Users\Administrator>
```

HR NTFS and SMB permissions were assigned to the HR resource group.



hr.test01 reached the delete operation for HR-Test.txt.

```
PS C:\WINDOWS\system32>  get-childitem "\\FS01\HR"
get-childitem : Access is denied
At line:1 char:2
+  get-childitem "\\FS01\HR"
+  ~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : PermissionDenied: (\\FS01\HR:String) [Get-ChildItem], UnauthorizedAccessException
    + FullyQualifiedErrorId : ItemExistsUnauthorizedAccessError,Microsoft.PowerShell.Commands.GetChildItemCommand

get-childitem : Cannot find path '\\FS01\HR' because it does not exist.
At line:1 char:2
+  get-childitem "\\FS01\HR"
+  ~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (\\FS01\HR:String) [Get-ChildItem], ItemNotFoundException
    + FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand

PS C:\WINDOWS\system32> whoami
lab\finance.test01
PS C:\WINDOWS\system32>
```

finance.test01 used the full UNC path \\FS01\HR and was refused.

Ticket #2003 - Automatic department drive mapping

Users should receive the correct network drive when they sign in instead of typing a UNC path manually.

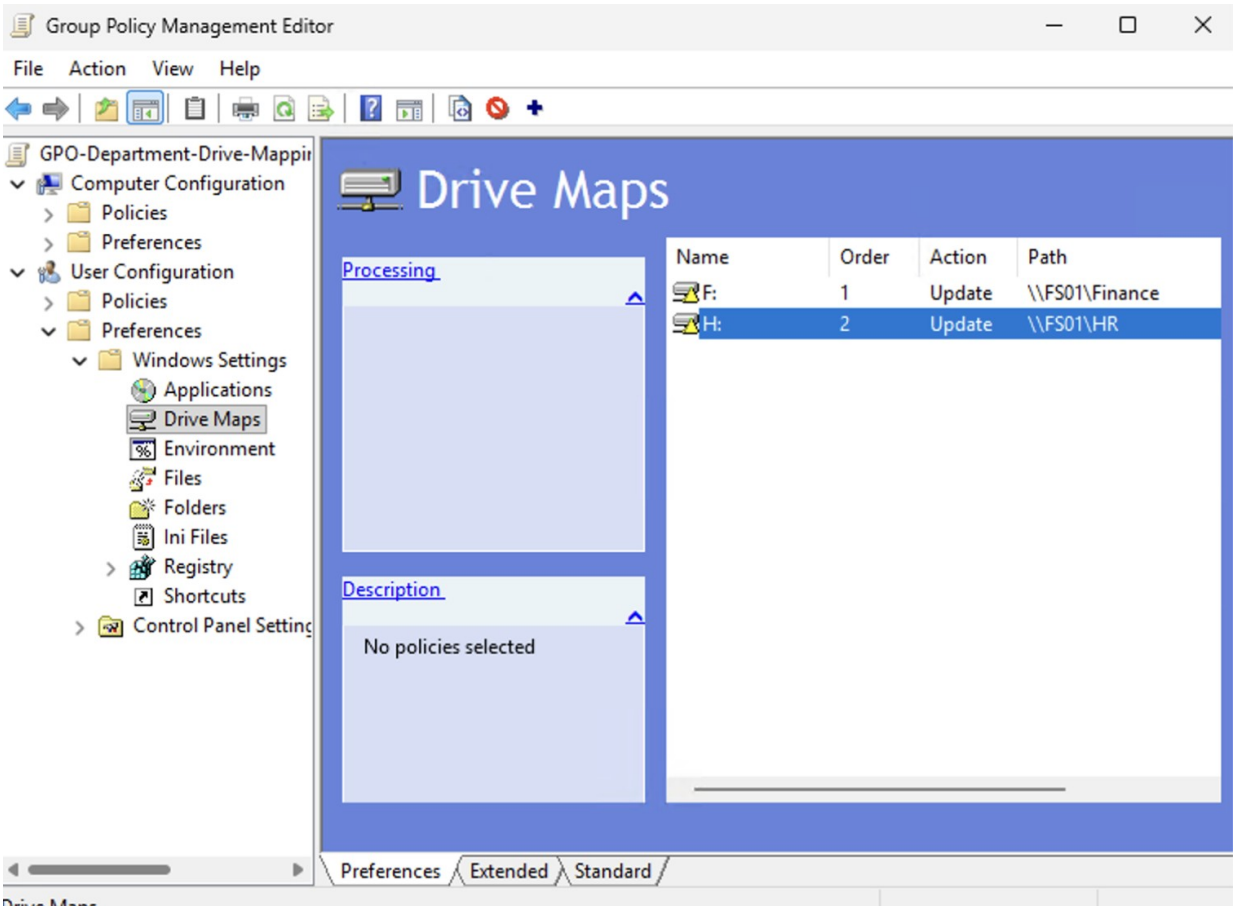
Request: Map the Finance and HR shares automatically for the correct users, while users with no department membership receive no department drive.

What I checked: I checked the GPO link on PortfolioLab, the Drive Maps entries, and the item-level targeting for Finance and HR.

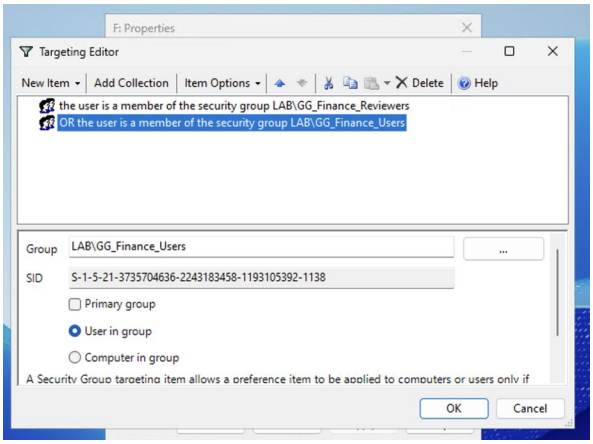
What I changed: I created F: for \\FS01\Finance and H: for \\FS01\HR. F: is targeted to GG_Finance_Users OR GG_Finance_Reviewers, so both modify users and read-only reviewers receive the same drive while the file ACL still controls what each group can do. H: is targeted to GG_HR_Users.

Issue encountered: I initially joined the two Finance targeting rules with AND. That would require a user to belong to both Finance groups, so normal Finance users would not receive F:. I changed the condition to OR before the final client test.

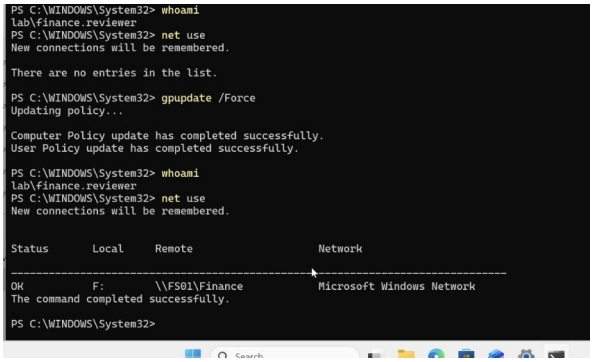
How I verified it: finance.test01 received F:, finance.reviewer received F:, hr.test01 received H:, and noaccess.test01 had no department drive. net use was used to verify the actual mapping on the client.



Drive Maps in GPO-Department-Drive-Mapping: F: points to Finance and H: points to HR.

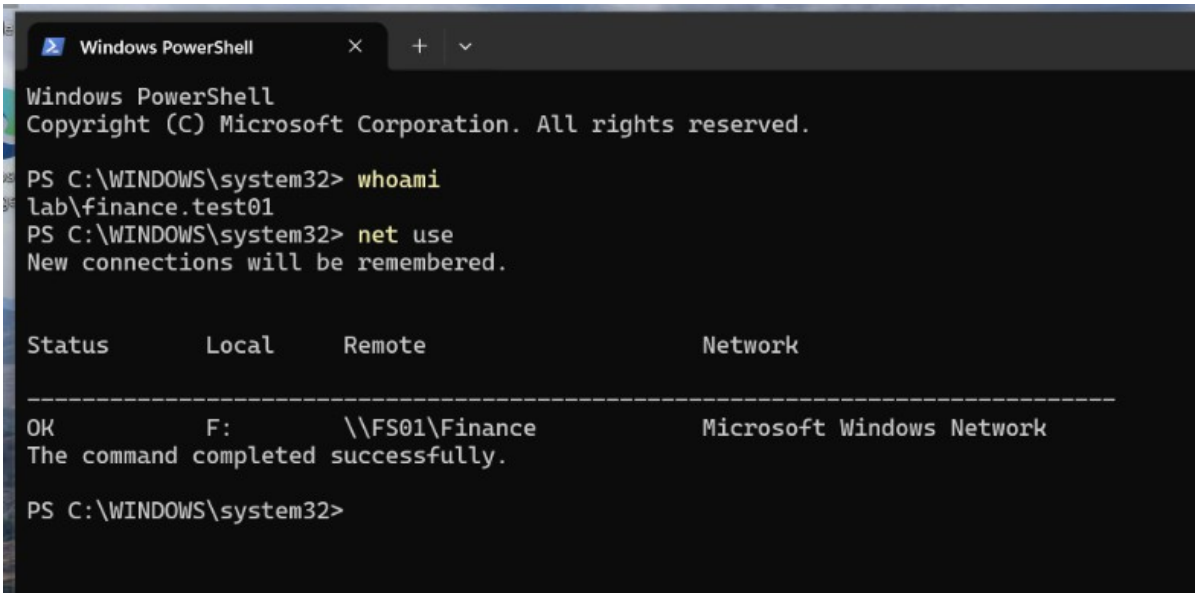


Finance item-level targeting accepts either Finance Users or Finance Reviewers.

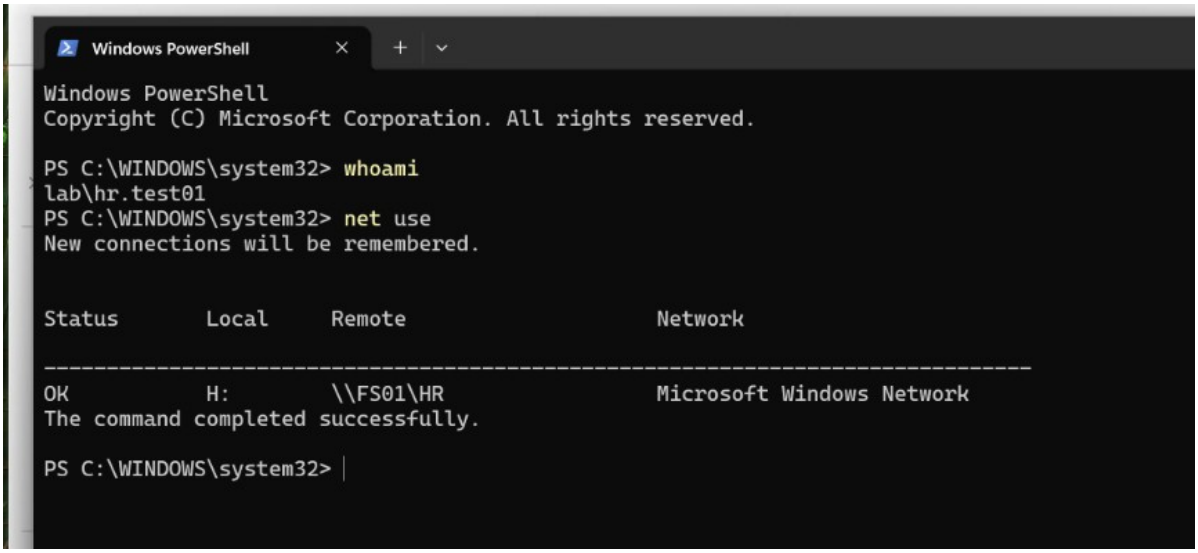


finance.reviewer receives F: after the updated policy is applied.

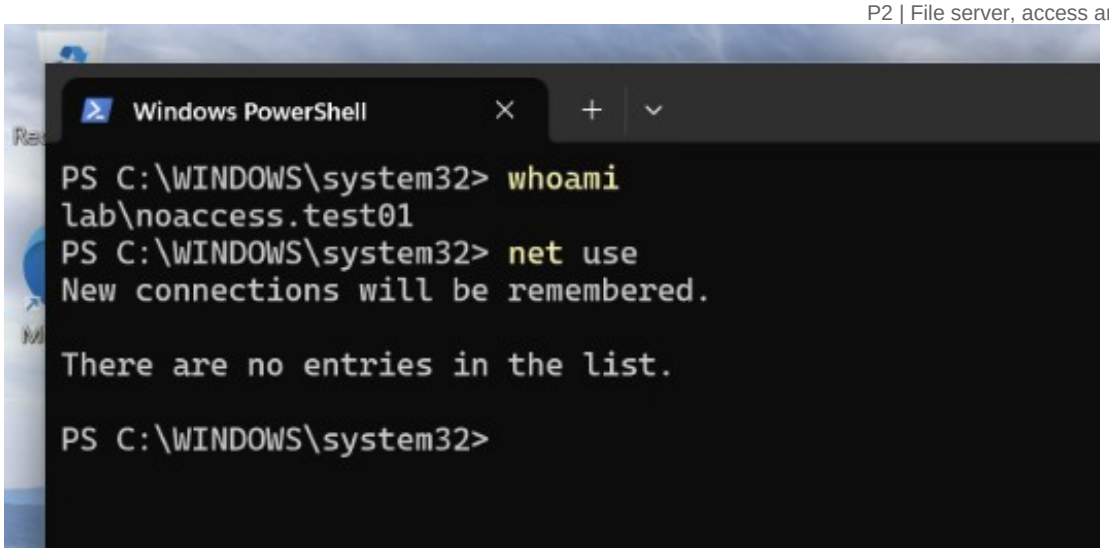
Ticket #2003 - client verification



finance.test01: net use shows F: mapped to \\FS01\Finance.



hr.test01: net use shows H: mapped to \\FS01\HR.



noaccess.test01: no mapped network drives are present.

Ticket #2004 - Move a user from Finance to HR

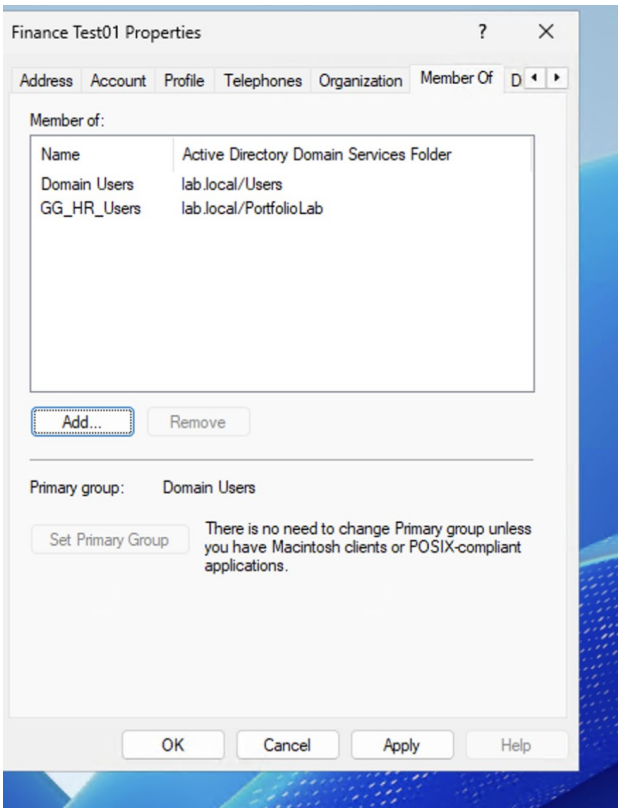
The account changes department and must lose Finance access while gaining HR access.

Request: Move finance.test01 from Finance to HR. The user must gain HR access, lose Finance access, lose the old Finance drive and have the AD account moved into the HR OU with the Department field updated.

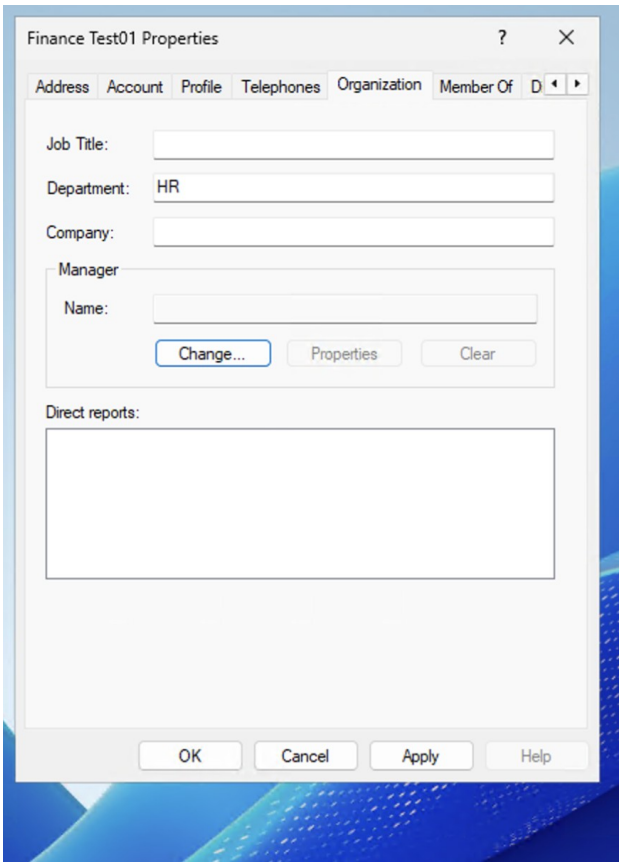
What I checked: I checked the current group memberships, the Department attribute, the OU location and the mapped drives visible in the user session.

What I did: I removed GG_Finance_Users, added GG_HR_Users, changed Department to HR and moved the same user object from PortfolioLab\Finance to PortfolioLab\HR. I did not change the username or password.

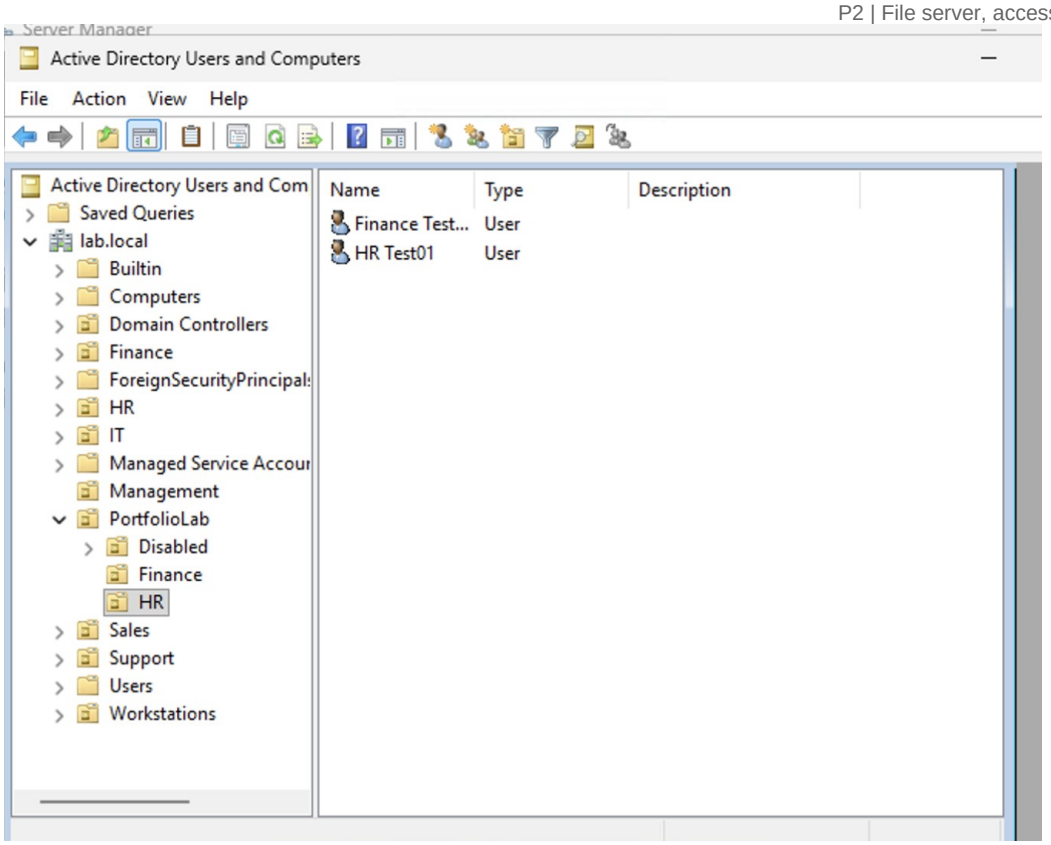
How I verified it: After a full sign-out and sign-in, the same account received H: to the HR share, F: was gone, \\FS01\HR opened, and \\FS01\Finance was refused. The fresh sign-in mattered because the existing Windows logon token still contained the old group membership.



Membership after the role change: Domain Users and GG_HR_Users.



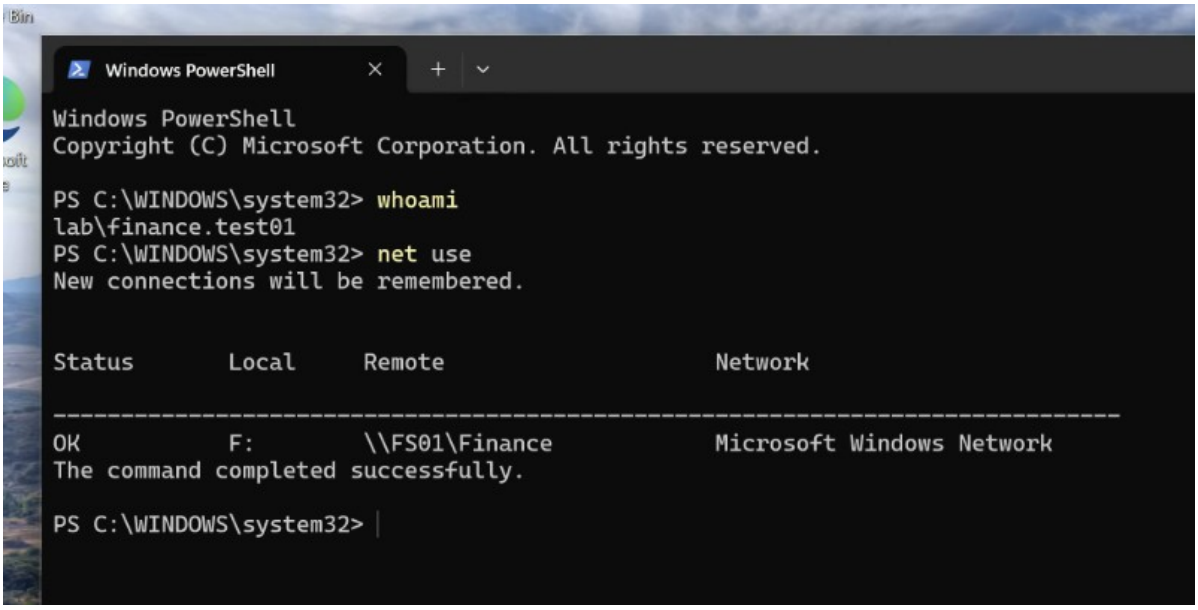
The Department attribute was updated to HR.



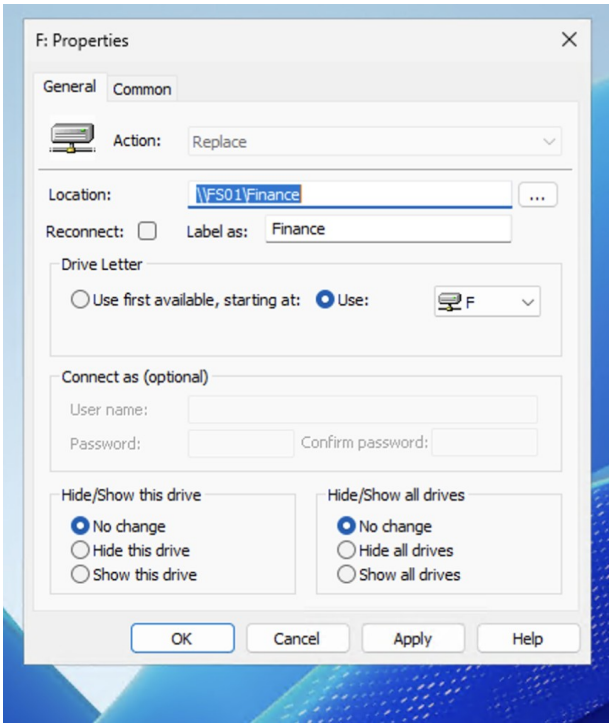
The same Finance Test01 account is now located in PortfolioLab\HR.

Ticket #2004 - stale drive and final result

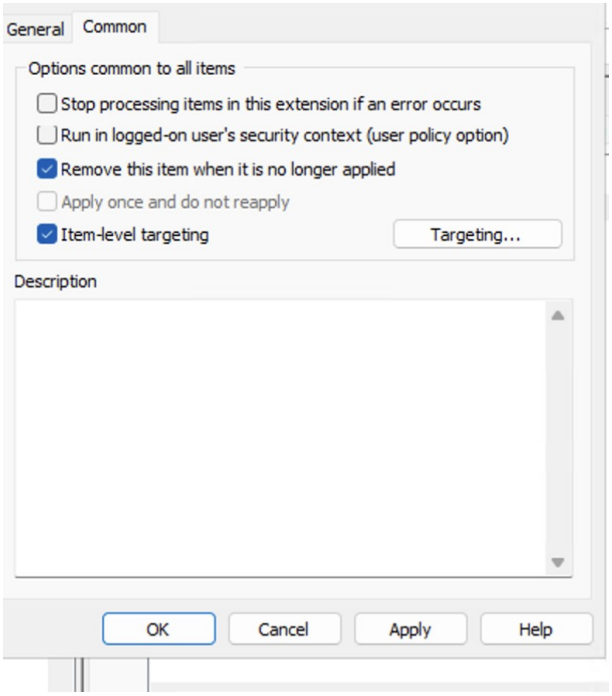
A useful problem appeared during the move. The old F: mapping remained after a policy refresh because the Finance drive was initially configured with Action = Update and the current session still carried the old Finance group token. I changed the Finance mapping to remove itself when it is no longer applicable; this changes the preference action to Replace. A full sign-out and sign-in then created a fresh security token.



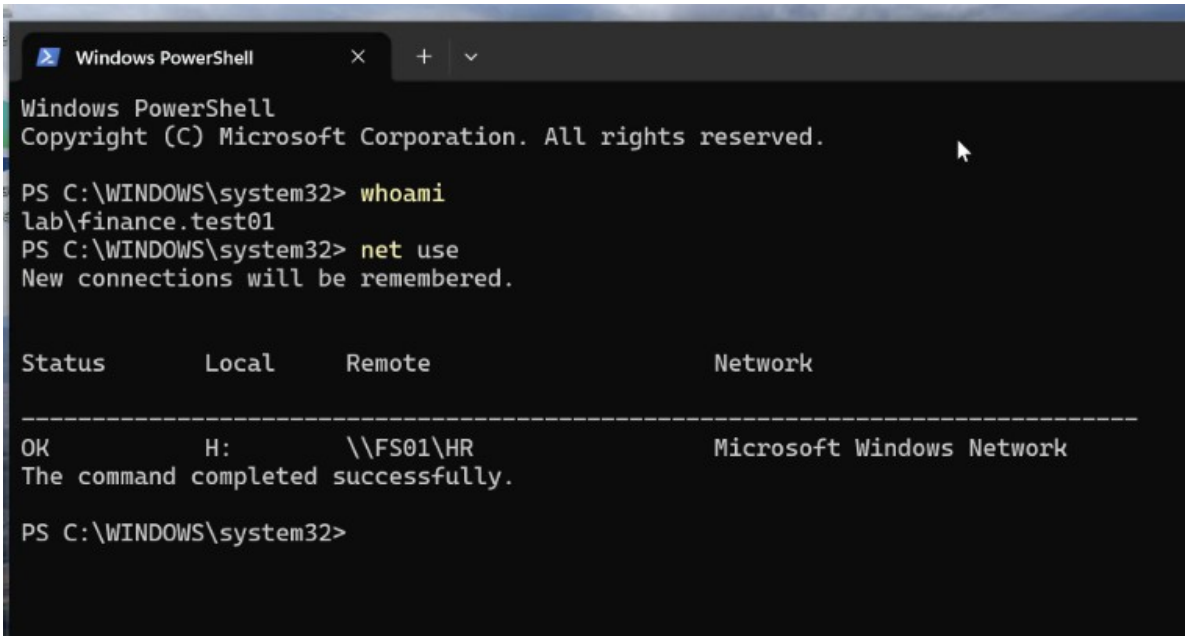
After the AD change, the current session still showed the old F: mapping.



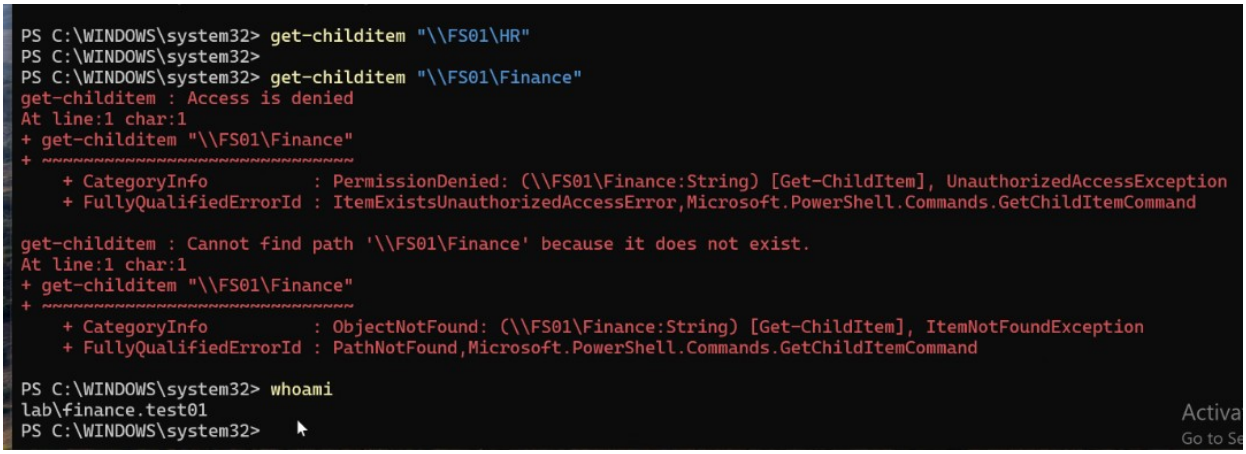
Finance mapping changed to Replace.



Remove this item when it is no longer applied is enabled.



After a fresh sign-in, finance.test01 now receives H: to \\FS01\HR and the old F: mapping is gone.



Direct access test: HR succeeds while the old Finance share is refused.

Ticket #2005 - Restore an accidentally deleted Finance file

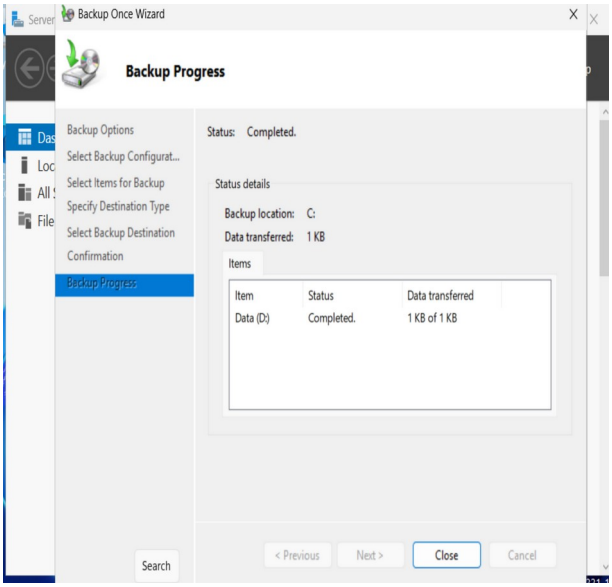
A backed-up budget file is deleted and then recovered to its original location with its permissions.

Request: Recover Budget-2026.txt after it is deleted from the Finance share. The restored file must contain the original text and retain the correct Finance ACL.

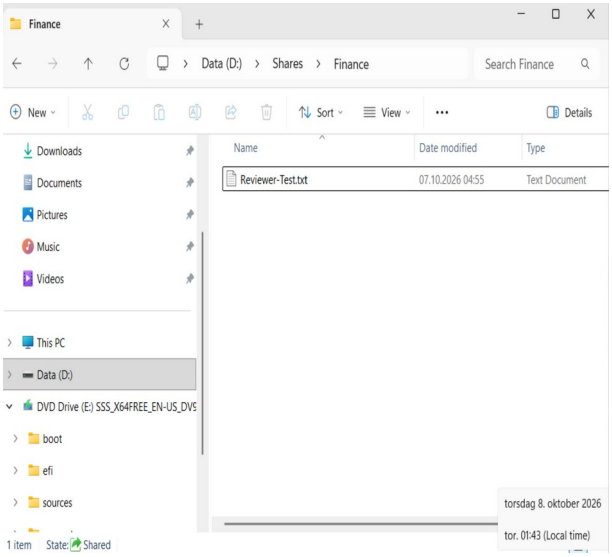
What I checked: Before deletion I verified the file content, the Finance ACL and that the backup had completed successfully.

What I did: I installed Windows Server Backup, created a custom backup of D:\Shares\Finance to Local Disk (C:), deleted Budget-2026.txt, and used Recover -> Files and folders to restore only that file to its original location with ACL recovery enabled.

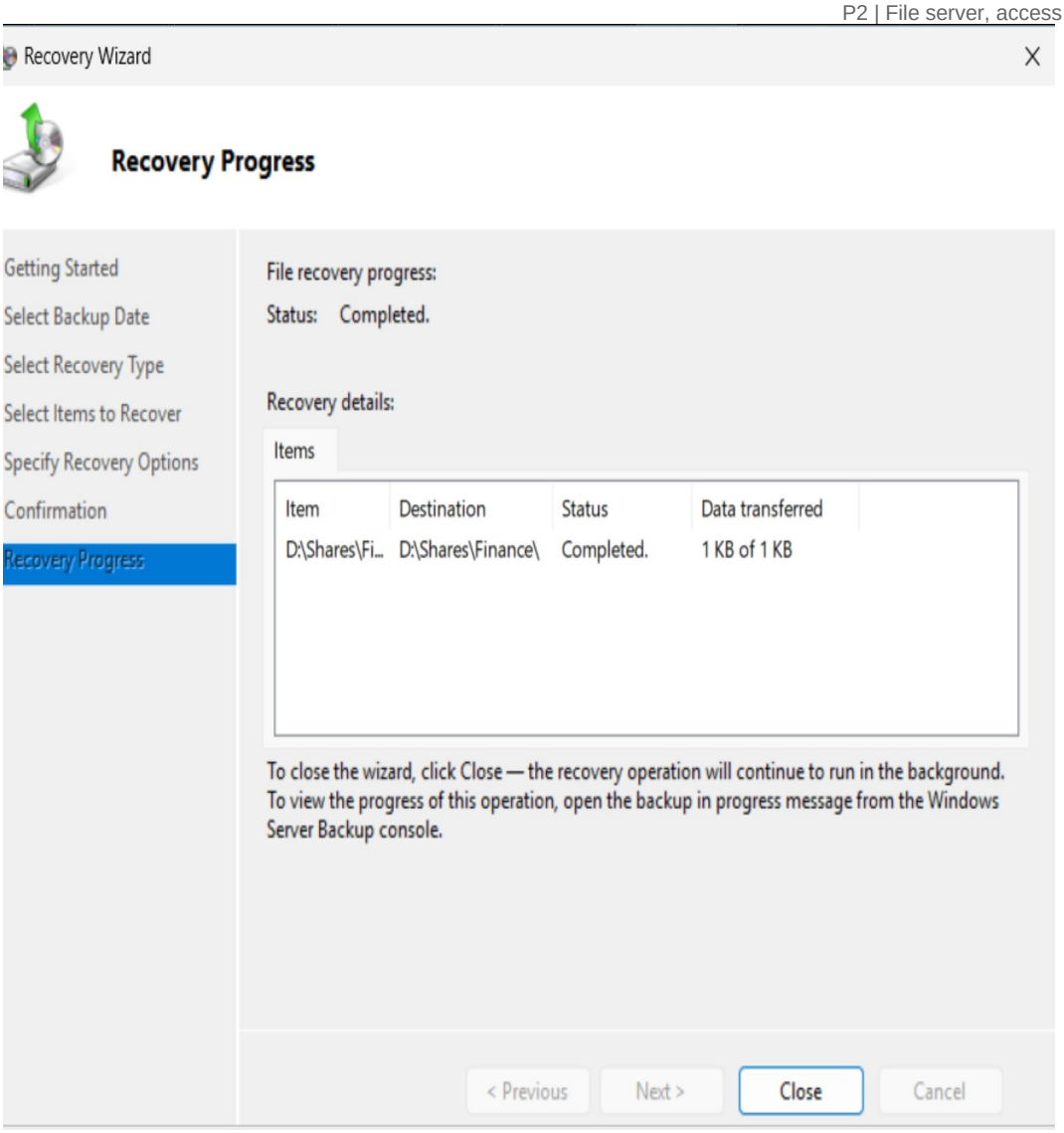
How I verified it: Windows Server Backup reported the recovery as completed. Budget-2026.txt reappeared in D:\Shares\Finance, Get-Content returned the original text, and icacls showed the same Finance Read and Modify groups plus Administrators and SYSTEM.



Windows Server Backup completed the Finance backup.



After the simulated accident, Budget-2026.txt is absent from the Finance folder.



File recovery completed to D:\Shares\Finance.

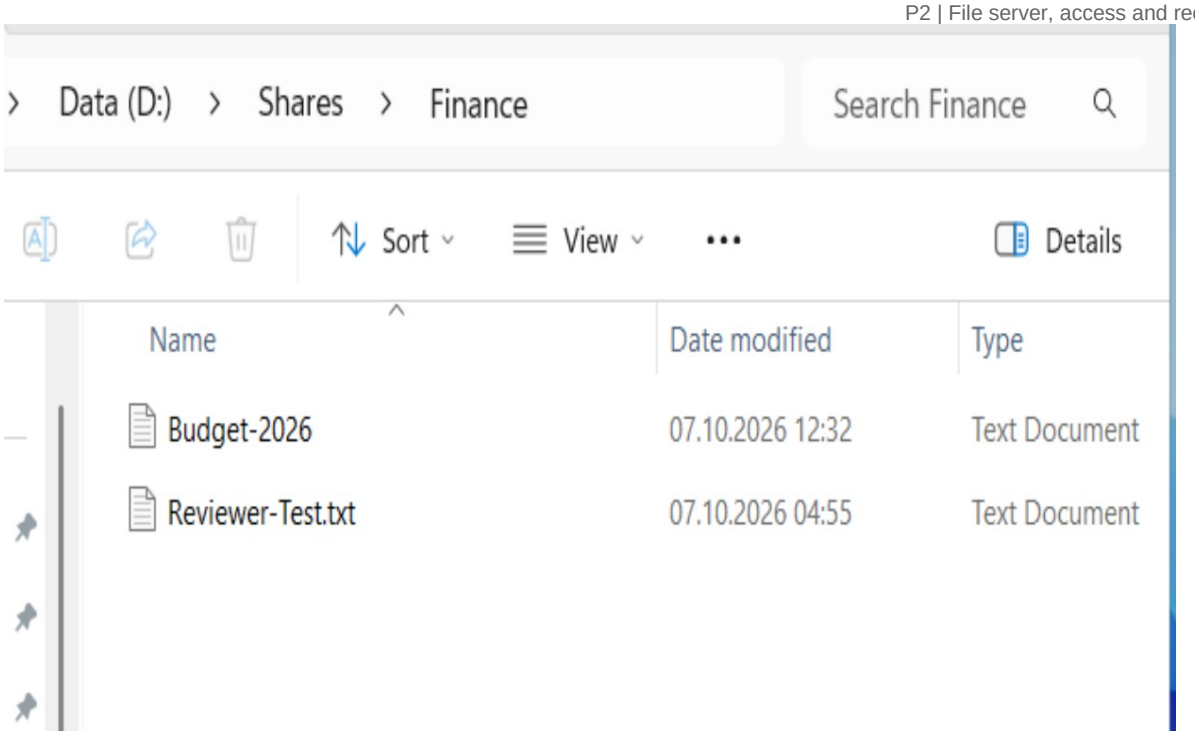
Ticket #2005 - restored content and permissions

```
PS C:\Users\Administrator> Get-Content "D:\Shares\Finance\Budget-2026.txt"
icacls "D:\Shares\Finance\Budget-2026.txt"
Finance Budget 2026 - TEST FILE
D:\Shares\Finance\Budget-2026.txt LAB\DL_FS01_Finance_Read:(I)(RX)
                                  LAB\DL_FS01_Finance_Modify:(I)(M)
                                  BUILTIN\Administrators:(I)(F)
                                  NT AUTHORITY\SYSTEM:(I)(F)

Successfully processed 1 files; Failed processing 0 files

PS C:\Users\Administrator>
```

The restored file contains the original text and the expected Finance ACL.



Final visual confirmation: Budget-2026 is back in the Finance folder.

Restore timing: about 11 minutes end to end in the recorded lab session, from deleting the file to a completed restore; the recovery copy itself took seconds. Backup limitation: this was a one-time copy stored on C: of the same FS01 VM. It demonstrates recovery from accidental deletion, but it does not protect against loss or compromise of the whole VM or host, ransomware affecting the server, or changes made after that backup. A production design would use scheduled backup to independent storage, with an offline or off-site copy.

Ticket #2006 - Group membership changed but access is still denied

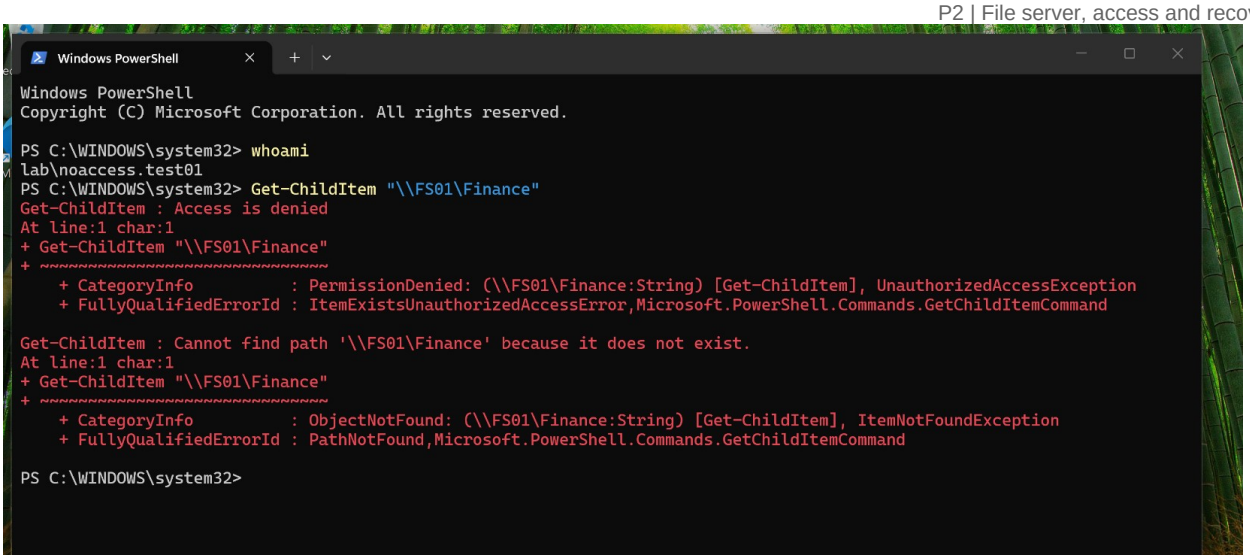
A user is added to the right AD group while already logged in, but the current Windows session still uses the old security token.

Request: Reproduce a case where IT adds a user to the correct group but the user still receives Access is denied. Find the cause and fix it without changing the folder permissions.

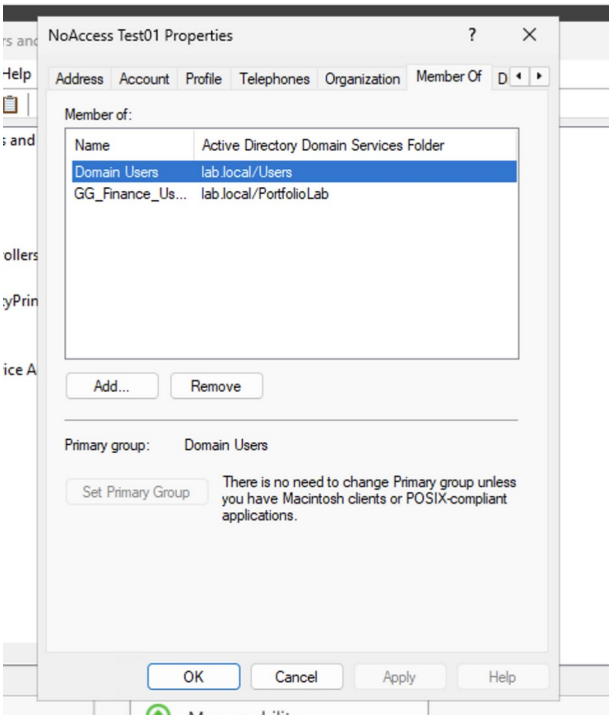
What I checked: I logged noaccess.test01 into WIN11-IT-01 first and confirmed that the account could not open \\FS01\Finance. While that same session remained active, I added the account to GG_Finance_Users on DC01.

What I did: I did not touch the FS01 ACL. In the existing session, whoami /groups still did not show a Finance group and the Finance share remained denied. I then signed the user out completely and signed back in.

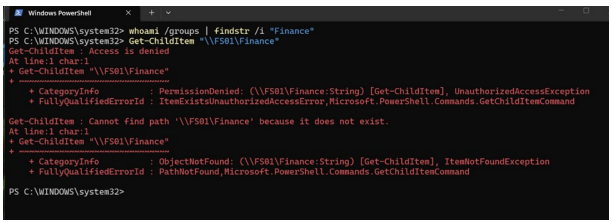
How I verified it: After the fresh sign-in, whoami /groups showed GG_Finance_Users and DL_FS01_Finance_Modify, and Get-ChildItem \\FS01\Finance listed the Finance files. The folder ACL never changed; the difference was the new Windows logon security token.



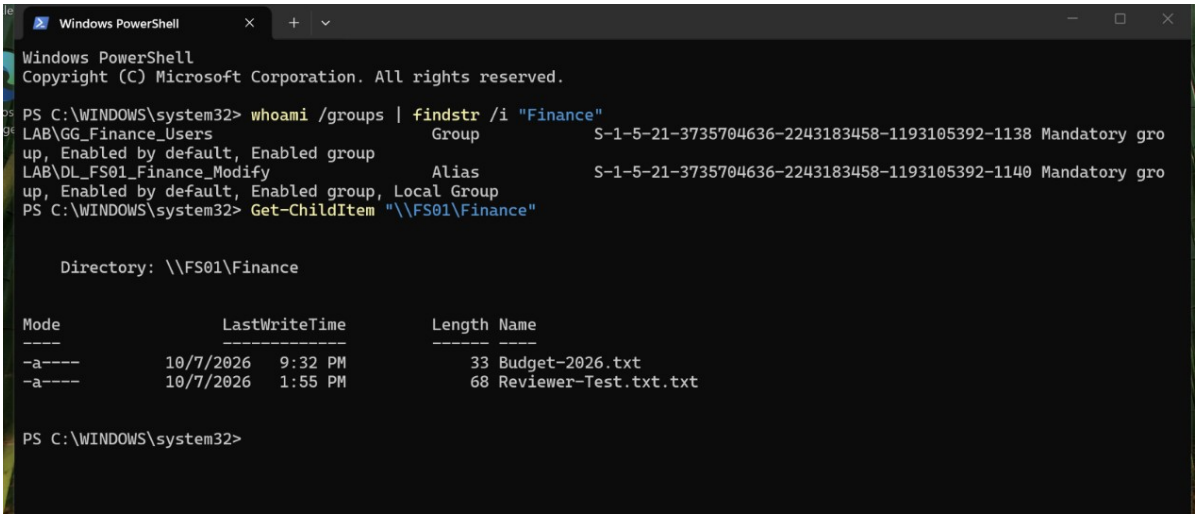
Before the group change, noaccess.test01 is refused at the Finance share.



DC01 shows noaccess.test01 added to GG_Finance_Users.



The already-running session still lacks Finance in its token and access remains denied.



After sign-out and sign-in, the token contains the Finance groups and the share opens successfully.

Project summary

The six tickets form one connected file-server administration exercise rather than six isolated tasks. The same group model is used to build Finance and HR access, map the right drives, move a user between departments, recover a deleted file, and troubleshoot a user whose permissions changed while the session was already active.

What this project demonstrates

- Windows Server member-server administration and SMB shares.
- Active Directory security groups and an AGDLP-style permission model.
- NTFS permissions, share permissions and effective access.
- Group Policy Preferences drive mapping with item-level targeting.
- A department transfer where old access is removed and new access follows the user.
- Windows Server Backup file-level recovery with ACL restoration.
- Troubleshooting of UNC path errors, Group Policy targeting, stale drive mappings and stale user security tokens.

Problems I had to solve

I kept the failures that changed how I understood the system: a malformed UNC path that looked like a network problem, an AND/OR targeting mistake in Group Policy Preferences, a stale F: drive during a department move, and a user session that still carried old group membership. In each case I checked the actual cause before changing the configuration.

What I learned and where the lab stops

The main value of the project was not only getting the shares to work, but learning which layer to check when they did not.

- Check inherited NTFS permissions before adding new access. A broad inherited entry can defeat an otherwise careful group design.
- Put groups on resources, not individual users. The Finance-to-HR move was handled by changing group membership instead of rewriting folder ACLs.
- Read the error before troubleshooting it. Cannot find path is a naming/path problem; Access is denied is a permission result.
- gpupdate refreshes policy, but it does not rebuild the user's Windows logon security token. Some group changes need a full sign-out and sign-in.
- Group Policy Preferences logic matters. AND versus OR changes who matches the rule, and Update versus Replace changes whether an old drive is removed.
- A backup is only proven by a restore. I verified the recovered file by comparing both its content and its ACL after recovery.

Lab limitations

- One domain controller and one file server: there is no server redundancy.
- Windows Server is an evaluation installation and lab.local is an internal lab domain.
- The Windows Server Backup copy is on the same FS01 VM and is not scheduled or off-site.
- The project proves administration and troubleshooting in a controlled lab; it is not presented as a production architecture.

The screenshots in this report are lab evidence. No passwords are shown. The environment is a personal VMware lab and is not presented as a production deployment.